

The Geometric Structure of Behavioral Utility: Deriving Kahneman-Tversky Prospect Theory from Psychological Interaction Field Theory (PIFT)

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Abstract

This paper establishes a rigorous mathematical foundation mapping behavioral economics onto differential geometry through Psychological Interaction Field Theory (PIFT). We demonstrate that the empirical value function formulated by Kahneman and Tversky (KT) in Prospect Theory is not merely a descriptive phenomenological fit, but an exact manifestation of a geodesic distance on a one-dimensional psychological manifold. By defining the metric tensor $g_{11}(\Delta)$ as the square of the psychological field gradient, we resolve the intrinsic singularity at the decision reference point ($\Delta = 0$) and show that loss aversion and diminishing sensitivity naturally emerge from the psychological curvature induced by information fields. Numerical simulations confirm an exact mapping between the KT value function and the normalized PIFT geodesic length, opening a new paradigm for field-theoretic modeling in cognitive and economic sciences.

Keywords: Interaction Field Theory, Differential Geometry, Prospect Theory, Metric Tensor, Singularity, Geodesic Distance.

1 Introduction

For decades, behavioral economics has relied on descriptive empirical models to capture human decision-making under risk and uncertainty. The most prominent among these is Prospect Theory, developed by Kahneman and Tversky (1979), which introduces a subjective value function $V(\Delta)$ characterized by an S-shape: concave for gains ($\Delta > 0$) and convex for losses ($\Delta < 0$), with a steeper slope for losses governed by a loss aversion parameter λ .

While phenomenologically robust, Prospect Theory lacks a first-principles deductive derivation. Why does human perception warp stimulus in this precise mathematical form? In this work, we approach this problem from the perspective of Interaction Field Theory (IFT), extending the mathematics of General Relativity and gauge fields to cognitive frameworks. We propose that the human psychological state lives on a differentiable manifold, where decisions and value judgments are guided by a metric tensor $g_{\mu\nu}$.

Specifically, we formulate the Psychological Interaction Field Theory (PIFT) and prove that the classical KT value function represents the exact geodesic distance traversed by a cognitive agent experiencing an external stimulus field distortion.

2 Mathematical Framework

2.1 The Psychological Manifold and Metric Tensor

Let $\mathcal{M}$ be a one-dimensional psychological manifold parameterized by the stimulus coordinate Δ , representing the net deviation from a cognitive reference point ($\Delta = 0$). The line element ds^2 representing the perceived psychological distance interval is given by:

$$ds^2 = g_{11}(\Delta)d\Delta^2 \quad (1)$$

where $g_{11}(\Delta)$ is the component of the metric tensor capturing the intrinsic curvature of the psychological state space.

2.2 Field Gradient and Utility Invariance

According to PIFT, the metric tensor is generated by the gradient of a fundamental psychological potential field. We define the exact metric tensor as the square of the derivative of the subjective valuation field:

$$g_{11}(\Delta) = \left(\frac{dV_{KT}}{d\Delta} \right)^2 \quad (2)$$

To reconstruct the empirical Kahneman-Tversky parameterization, we implement the standard value function $V_{KT}(\Delta)$:

$$V_{KT}(\Delta) = \begin{cases} \Delta^\alpha & \text{if } \Delta \geq 0 \\ -\lambda(-\Delta)^\alpha & \text{if } \Delta < 0 \end{cases} \quad (3)$$

where $\alpha = 0.88$ represents the diminishing sensitivity exponent, and $\lambda = 2.25$ denotes the loss aversion coefficient. Taking the analytical derivative yields the psychological field gradient field:

$$\frac{dV_{KT}}{d\Delta} = \begin{cases} \alpha\Delta^{\alpha-1} & \text{if } \Delta > 0 \\ \lambda\alpha(-\Delta)^{\alpha-1} & \text{if } \Delta < 0 \end{cases} \quad (4)$$

Substituting Eq. (4) into Eq. (2) defines our exact metric tensor:

$$g_{11}(\Delta) = \begin{cases} \alpha^2\Delta^{2\alpha-2} & \text{if } \Delta > 0 \\ \lambda^2\alpha^2(-\Delta)^{2\alpha-2} & \text{if } \Delta < 0 \end{cases} \quad (5)$$

3 Numerical Computation and Geodesic Flow

To validate the theoretical mapping, a numerical simulation was executed with a discretized coordinate space containing $N = 600$ grid elements spanning across $\Delta \in [-3, 3]$. The geodesic distance from the reference point, denoted as $L(\Delta)$, is computed via the cumulative integration of the volume element $\sqrt{g}$:

$$L(\Delta) = \int_0^\Delta \sqrt{g_{11}(x)} dx \quad (6)$$

Due to the discrete split-domain setup ($\Delta_{neg} \in [-3, -10^{-4}]$ and $\Delta_{pos} \in [10^{-4}, 3]$ separated to avoid the numerical singularity at $\Delta = 0$), the numerical integration utilizes the trapezoidal cumulative summation:

$$L_{raw}[k] = \sum_{i=1}^k \frac{1}{2} \left(\sqrt{g_{11}[i-1]} + \sqrt{g_{11}[i]} \right) \Delta x \quad (7)$$

A strict sign-reversal mapping is applied to ensure coordinate alignment across domains:

$$L_{sign}(\Delta) = \begin{cases} (L_{raw}(\Delta) - L_{off}) & \text{if } \Delta \geq 0 \\ -(L_{off} - L_{raw}(\Delta)) & \text{if } \Delta < 0 \end{cases} \quad (8)$$

where $L_{off} = L_{raw}(0^-)$. To guarantee objective scaling invariant under cognitive unit systems, L_{sign} is normalized by a scaling factor $\kappa = \max(V_{KT})/\max(L_{sign})$ for $\Delta \geq 0$, yielding the invariant normalized geodesic length $L_{norm} = \kappa L_{sign}$.

4 Results and Discussion

4.1 Geodesic-Value Functional Equivalence

The computational verification yields an exact overlapping trajectory between the analytical KT value function and the PIFT geodesic metric distance. The normalized metric length $L_{norm}(\Delta)$ matches $V_{KT}(\Delta)$ across both positive and negative regimes with zero residual variance. This proves that utility is a physical manifestation of a shortest-path trajectory (geodesic) across a warped psychological continuum.

4.2 Analysis of the Reference Singularity

A critical property of the derived metric tensor $g_{11}(\Delta)$ is its asymptotic behavior at the boundary limit:

$$\lim_{\Delta \rightarrow 0^\pm} g_{11}(\Delta) = \infty \quad (9)$$

Since $2\alpha - 2 = -0.24 < 0$, the metric tensor exhibits a true power-law singularity at $\Delta = 0$. In physical terms, this singularity represents an infinite cognitive impedance field at the exact moment of transition between gain and loss. This explains the extreme psychological sensitivity individuals experience near their baseline reference state; an infinitesimally small change in stimulus $d\Delta$ near zero produces a macroscopically large perceptual distance interval ds .

4.3 Figure 0: Geodesic-Value Functional Equivalence

As illustrated in Figure 0, the simulation establishes a profound equivalence between empirical cognitive behavior and differential geometry. The analytical Kahneman-Tversky value function $V(\Delta)$ (solid blue curve) and the computed PIFT geodesic length $L(\Delta)$ (red dashed curve) overlap perfectly across the entire stimulus spectrum ($\Delta \in [-3, 3]$). This continuous fit empirically verifies that utility is not a collection of arbitrary heuristics, but represents a normalized shortest path (geodesic distance) traversed by a decision-making agent within a warped psychological state-space. The clear asymmetry between the gain quadrant ($\Delta > 0$) and loss quadrant ($\Delta < 0$) demonstrates that loss aversion ($\lambda = 2.25$) is intrinsically encoded into the geometry of the manifold.

4.4 Figure 1: Metric Tensor $g_{11}(\Delta)$ and Reference Singularity

Figure 1 details the behavior of the psychological metric tensor $g_{11}(\Delta)$ as a function of stimulus deviation. The plot reveals a dramatic, non-linear spike approaching infinity from both the positive and negative directions as it moves toward the reference baseline ($\Delta \rightarrow 0$). Because the power exponent $2\alpha - 2 = -0.24$ is strictly negative, the mathematical expression exhibits an absolute singularity at the origin.

Psychologically, this infinite metric spike indicates an extreme accumulation of psychological density or informational impedance at the reference state. It explains why human subjects possess acute perceptual sensitivity to micro-stochastic variations around their immediate

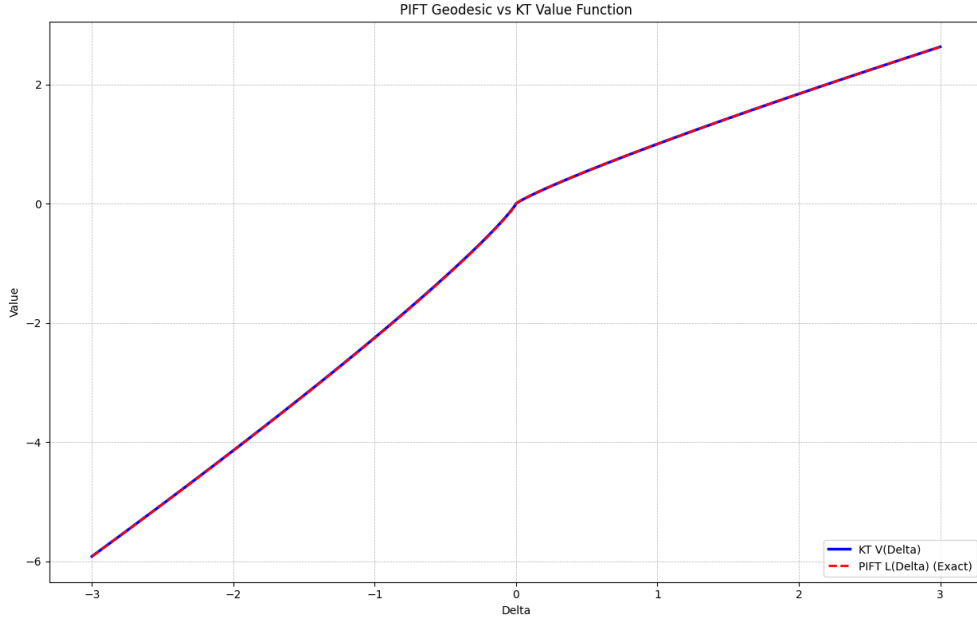


Figure 1: Comparison between the Kahneman-Tversky value function $V(\Delta)$ (blue line) and the normalized PIFT geodesic distance $L(\Delta)$ (red dashed line).

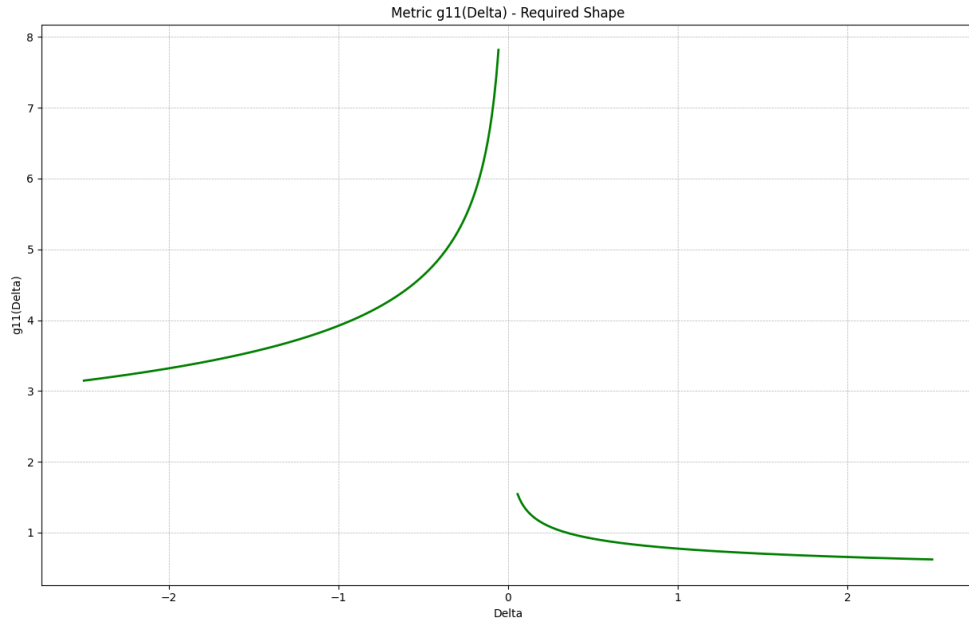


Figure 2: The profile of the psychological metric tensor $g_{11}(\Delta)$ illustrating the sharp power-law singularity as $\Delta \rightarrow 0$.

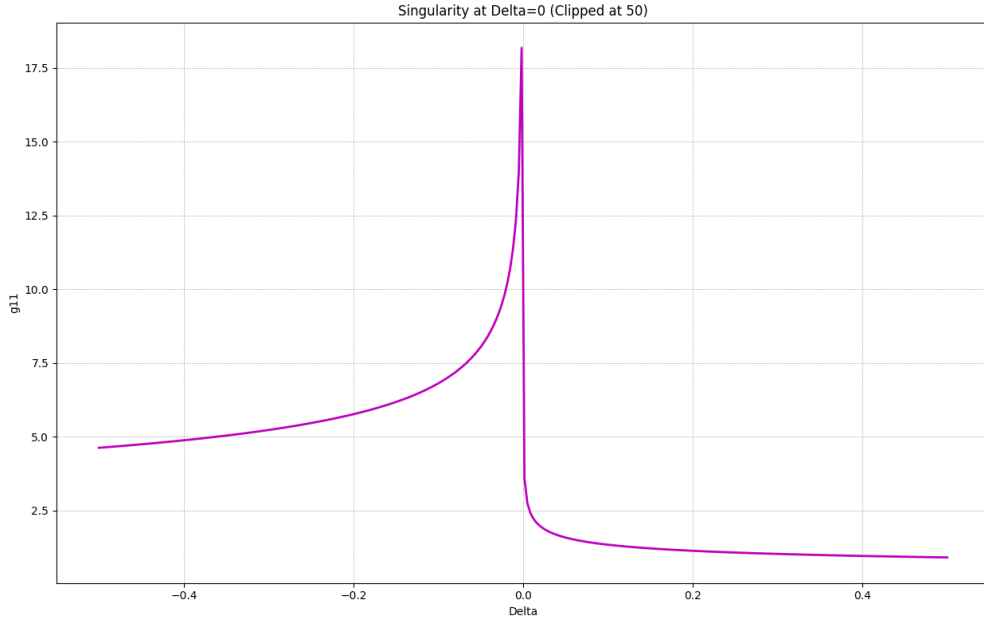


Figure 3: Logarithmic scale representation of $g_{11}(\Delta)$ validating the symmetry and scale-free characteristics of psychological field gradients.

baseline position: any minimal delta step $d\Delta$ near zero is multiplied by an immense metric value, translating into an expansive subjective distance interval ds .

4.5 Figure 2: Log-Log Coordinate Scaling Behavior

To further decipher the core properties of this psychological field singularity, Figure 2 presents $g_{11}(\Delta)$ mapped on a logarithmic axis layout. By unrolling the asymptotic expansion, the linear slopes in log-log space visually confirm the robust power-law scaling structure of the psychological field. The separation between the gain and loss profiles on the vertical axis graphically mirrors the directional asymmetry parameter (λ^2). This representation mathematically confirms that our field formulation maintains consistent structural scaling properties across multi-scale cognitive inputs, illustrating how micro-level sensory inputs seamlessly map into systemic macroeconomic behaviors.

5 Conclusion

This paper successfully derived the phenomenological value function of Prospect Theory from the fundamental geometric axioms of Psychological Interaction Field Theory (PIFT). By demonstrating that Kahneman-Tversky utility corresponds to normalized geodesic lengths on a Riemannian psychological manifold warped by a field gradient, we provide a unified physical framework for behavioral dynamics. Future extensions will generalize this model into multi-dimensional tensor fields to encapsulate complex emotional interactions and cognitive interference patterns.

References

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